

On Equalized Odds in Supervised Learning, for the Special Case of Non-Decreasing Conditional Event Probabilities *

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Abstract

Algorithmic fairness is an increasingly important aspect of algorithm design as algorithms play an increasing role in society. There are many competing notions of algorithmic fairness and here we consider a well studied notion called equalized odds. For a given classification problem, equalized odds requires the amount of false positive error to be equal across demographics and the amount of true positive error to be equal across demographics. Foundational work by Hardt, Price, and Srebro [13] considers the problem of taking an existing classifier or a rating system as a black box and deriving another classifier satisfying equalized odds and otherwise minimizing the error. They gave a polynomial time algorithm that produces a fairly simple classifier that is optimal (in a particular sense, with respect to a given loss function) among those that satisfy equalized odds.

In this work, we further the research direction of Hardt, Price, and Srebro [13] and in particular we consider the same problem for the special and canonical case of algorithmic scoring systems that exhibit *non-decreasing conditional event probabilities*. Non-decreasing conditional event probabilities is a natural property inherent to reasonable scoring systems. We show that for scoring systems with non-decreasing event probabilities the optimal derived classifier can always be obtained by an extremely simple, *randomized one-threshold classifier*, which involves only a single threshold for each demographic. Moreover, the optimal randomized one-threshold classifier can be computed efficiently. Given the ubiquity of non-decreasing conditional event probabilities, constructing such a radically simple mechanism to achieve equalized odds, that is also optimal among those that achieve equalized odds, is valuable from the perspective of transparency. It gives interesting structural insight into equalized odds in general, which we also discuss.

1 Introduction

Algorithms have increasingly assumed an integral part of society, making or driving decisions that significantly impact individual lives. Unfortunately, these algorithms can produce unequal outcomes for individuals. In domains like criminal justice, education, and employment, algorithms help make more accurate decisions, but can also perpetuate and exacerbate existing biases. This calls for a thoughtful and robust approach to enforcing *fairness* on algorithms.

An important example that has gained significant attention is the COMPAS risk tool made by Northpointe. Across the United States, COMPAS and other algorithms are used to aid decisions in the criminal justice system. COMPAS scores, specifically, are used by judges and parole officers to predict criminal recidivism, which is defined as the likelihood of a criminal to re-offend. William Dieterich and Brennan [23] from the

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Northpointe research department show that the decisions made by COMPAS risk tool are *well-calibrated*. In this context, well-calibrated means that, within each demographic, the total number of recidivists predicted by COMPAS (closely) matches the true number of recidivists. While the decisions made by the COMPAS risk tool are well-calibrated, Angwin et al. [1] found that the COMPAS risk tool makes different kinds of errors that are unequally distributed across demographics in the following sense. [1] noted that “black defendants were far more likely than white defendants to be incorrectly judged to be at a higher risk of recidivism” (i.e., there is a higher *false positive rate* for the Black demographic). Additionally, “white defendants were more likely than black defendants to be incorrectly flagged as low risk”; (i.e., there is a higher *false negative rate* for the white demographic). The point is that the *false positives* and *false negatives* were unfairly distributed. Meanwhile, COMPAS risk scores are calculated based on criminal history, job status, level of education and family information. These features don’t explicitly identify an individual’s race. However, these features can be implicitly correlated with race, and the correlations leak into the biased outcomes.

Another important example is FICO scores. FICO scores are the industry standard in the United States for making loan decisions. Like COMPAS scores, FICO scores do not explicitly take into account an individual’s race. They are calculated based on features like the number of bank accounts an individual has, but these features could be correlated with race in subtle ways. Hardt, Price, and Srebro [13] conducted a similar statistical analysis as [1] where they studied decision rules based on FICO scores. These rules predicted whether an individual will default on a loan. Hardt, Price, and Srebro [13] found that these decision rules based on FICO scores output a higher number of false positives in the Black demographic than the White or Asian demographic, which is unfair.

In addition to these real world examples there are several natural reasons as to why data driven algorithms can produce unfair outcomes. To some extent, machine learning models are only as fair as the data they use, as machine learning algorithms are designed to learn patterns from data to make decisions. Some patterns provide appropriate and useful information, while other patterns represent stereotypes that we would prefer to suppress. Unfortunately, machine learning models cannot distinguish undesirable stereotypes from other patterns, which leads to unfair outcomes [2]. Another liability is sample size disparity, defined as the difference between the number of training samples from each demographic. In the presence of sample size disparity, a machine learning model learns better for demographics with more training samples, creating disparate outcomes. For example, algorithms in healthcare may produce less effective treatment plans for groups that are poorly represented in the data.

Since algorithms play a significant role in people’s lives, it is important to understand how these algorithms make decisions. But in the pursuit of low prediction error, algorithms can develop very complicated strategies. For example, neural networks have a higher predictive power but also a more complex and opaque structure compared to simple statistical methods. There is recent interest in improving the explainability and interpretability of these recent techniques, which could help identify and address the bias that can arise [3, 4, 11]. This paper focuses on another approach based on post-processing the output of existing algorithms towards achieving some measure of fairness. A driving motivation of our work is the importance of transparency in such interventionist mechanisms.

1.1 Theoretical frameworks for fairness

Having discussed some real world scenarios and inherent reasons for algorithmic bias, we review the existing foundational literature that formulates different notions of fairness. One criteria for fairness is called **demographic parity**. Demographic parity requires the algorithm’s prediction for an individual to be statistically independent of their demographic. In the case of binary classification, demographic parity implies that the percentage of positive labels for each demographic is equal. Another approach to fairness is **well-calibration**, as discussed earlier. Well calibration is also referred to as **calibration within groups**. Calibration within groups means that within each demographic, the expected number of positive instances

predicted by an algorithm (closely) matches the number of true positive instances. Another notion of fairness is called **equalized odds**. Equalized odds forces *true positive rates* and *false positive rates* of a binary classifier, conditional on the demographic, to be equal across all demographics. There is a similar but weaker notion to equalized odds called **equalized opportunity**. Equalized opportunity means that (just the) conditional true positive rates of a binary classifier is equal across demographics. We note that there are additional notions of fairness not discussed in detail here; see (e.g.) [5, 16, 18, 19].

Dwork et al. [7] argues that demographic parity is an inadequate notion of fairness for several reasons. First, a demographic might be a helpful feature for predicting the outcome. The assertion that there is no correlation between the demographic and outcome reduces the accuracy of the algorithm, sometimes to a crippling extent. Demographic parity can also be unfair in the following sense. Consider a scenario suggested by [13], in which candidates are being selected from applications from two demographics, A and B. Suppose the demographic A has more qualified applicants than demographic B. To make the percentages of positive labels in both the demographics match, demographic parity will force the algorithm to accept unqualified individuals demographic B. If an employer has a cap on the total number of new hires, then qualified individuals from demographic A will be rejected to allow unqualified individuals from demographic B to be hired. In turn, unqualified individuals from demographic B can perform poorly and mislead the employer to avoid hiring from that demographic in the future.

While any reasonable definition for fairness is generally desirable, different definitions can conflict with one another in surprising ways. Kleinberg, Mullainathan, and Raghavan [15] showed that calibration within groups and equalized odds cannot be achieved simultaneously unless there is either **perfect prediction** or **equal base rates** across demographics. Perfect prediction is when an algorithm makes no prediction error, and equal base rates is when all the demographics have the same proportion of members in the positive class. Most real world examples don't satisfy either of these conditions. In the earlier examples of COMPAS scores, neither perfect prediction or equal base rates are satisfied. Given these facts, the report by [23] that COMPAS score is well-calibrated *implies* that COMPAS scores do not satisfy equalized odds, as observed by [1].

1.2 The HPS framework for achieving equalized odds

In this work, we are particularly interested in the framework of Hardt, Price, and Srebro [13], applied to the equalized odds fairness criteria¹. We first give a more formal definition of equalized odds. We assume a supervised learning setting where there are true outcomes $Y \in \{0, 1\}$ and a set of features X . We also assume another categorical variable $A \in \mathcal{A}$. For example, $\mathcal{A}$ could be a set of different ethnicities, or the set of different gender denominations. The labeled training data induces a joint distribution (Y, A, X) .

Let $\tilde{Y}(X, A) \in \{0, 1\}$ be a binary classifier. Then $\tilde{Y}(X, A) \in \{0, 1\}$ satisfies **equalized odds** (with respect to demographics $A \in \mathcal{A}$ and true outcomes $Y \in \{0, 1\}$) if $\tilde{Y}$ and A are independent conditional on Y . In other words, a predictor $\tilde{Y}$ satisfies equalized odds if and only if the following holds for all $a, a' \in A$.

$$\begin{aligned}\mathbf{P}[\tilde{Y} = 1 \mid Y = 0, A = a] &= \mathbf{P}[\tilde{Y} = 1 \mid Y = 0, A = a'] \\ \mathbf{P}[\tilde{Y} = 1 \mid Y = 1, A = a] &= \mathbf{P}[\tilde{Y} = 1 \mid Y = 1, A = a']\end{aligned}$$

The first equality says that the false positive rates conditional on the demographic are equal across all demographics. The second equality says that the true positive rates conditional on the demographic are equal across all demographics. Of course, this also implies that the conditional true negative and false negative rates are also equal across all demographics. Note that equalized odds allows the classifier $\tilde{Y}(R, A)$ to have some limited dependency on demographic A . Consequently, requiring equalized odds does not (necessarily, and empirically) hamper the accuracy of the predictor $\tilde{Y}(R, A)$ to the same extent as demographic parity.

¹[13] also applies their framework to equalized opportunity, and comes to many similar conclusions for equalized opportunity as with equalized odds.

Many of the examples discussed earlier were real-valued scores R that are then converted to a (possibly randomized) binary classifier $\tilde{Y}(R, A) \in \{0, 1\}$. Hardt, Price, and Srebro [13] consider the problem of taking as input a distribution (Y, A, R) over true outcomes $Y \in \{0, 1\}$, scores $R \in \mathbb{R}$ and demographics $A \in \mathcal{A}$, and computing an optimal $\tilde{Y}(R, A) \in \{0, 1\}$ subject to $\tilde{Y}(R, A)$ satisfying equalized odds. Here “optimality” is measured by a linear loss function that penalizes false positive and false negative errors, and (loosely speaking) is relative to other classifiers based on thresholding rules. To clarify this latter point, we first formally define deterministic and randomized threshold classifiers, as follows.

Definition 1. Let $R \in \mathbb{R}$ be as random variable. A **deterministic threshold classifier** $I_t : \mathbb{R} \rightarrow \{0, 1\}$ consists of a real value $t \in \mathbb{R}$, and is defined by

$$I_t(R) = \begin{cases} 1 & \text{if } R \geq t \\ 0 & \text{otherwise.} \end{cases}$$

Beyond deterministic threshold classifiers, we also consider *randomized* thresholds. At a high level, a randomized threshold classifier applies a threshold classifier randomly selected from some distribution of thresholds. In this work, we are interested in the number of threshold classifiers in the support of this distribution, as a measure of complexity of the randomized threshold classifier. The formal definition is as follows.

Definition 2. Let (A, R) be jointly distributed random variables with $A \in \mathcal{A}$ for a finite set $\mathcal{A}$ and $R \in \mathbb{R}$. Let $Y(R, A) \in \{0, 1\}$ be a classifier derived from (R, A) . For $k \in \mathbb{N}$, we say that $Y(R, A)$ is a **randomized k -threshold classifier** if for each $a \in \mathcal{A}$, there are k probabilities $p_{a,1}, \dots, p_{a,k} \in [0, 1]$ and k thresholds $t_{a,1}, \dots, t_{a,k} \in \mathbb{R}$ with the following properties.

1. For all $a \in \mathcal{A}$, $p_{a,1} + \dots + p_{a,k} \leq 1$.
2. For all $A \in \mathcal{A}$, conditional on $A = a$, $Y(R, A)$ is equivalent to the following randomized algorithm. Randomly select either an index $i \in [k]$ or no index at all, where the probability that i is selected is $p_{a,i}$. If an index $i \in [k]$ is selected, then apply the deterministic threshold I_{t_a} to R . If no index i is selected, return 0.

The algorithm in [13] produces a randomized 2-threshold classifier, where the probabilities within each demographic $a \in \mathcal{A}$ sum to 1. That is, for an individual from a fixed demographic $a \in \mathcal{A}$, this classifier randomly selects one of two deterministic thresholds and compares it to the individual’s score. One can argue that the randomized 2-threshold classifier produced by the algorithm is optimal compared to any randomized k -threshold classifier (for any k). An important contribution of [13] is to establish a connection between equalized odds (as well as equal opportunity) and randomized threshold classifiers, which leads to the algorithmic results mentioned above. We also note that the algorithm of [13] does not revisit the original features X or consider the mechanics underlying the score R . It is purely a post-processing step that takes R as a black box, which makes it applicable to any existing rating system.

An important related work is by Corbett-Davies et al. [6], which also takes an optimization perspective on fairness although there are some differences in the setup. The objective function considered by [6] is qualitatively different from the type studied in [13]. The classifiers also implicitly assumes access to the optimal Bayes regressor (defined below) rather than arbitrary scoring functions.² That said, subject to each of several fairness conditions – including demographic, conditional demographic parity, and equal false positive rates (which is technically very similar to equal opportunity) – [6] showed that one can obtain deterministic threshold classifiers that are optimal among all classifiers, in a strongly universal sense. [6] also includes extensive discussion about the use of different types of threshold-based classifiers, some of which is relevant to this work.

²We suspect their framework can be extended to any scoring function with non-decreasing conditional event probabilities (defined below).

1.3 Equalized odds, non-decreasing conditional event probabilities, and one randomized threshold

Thus far we have introduced different notions of fairness and briefly explained the framework in [13] for equalized odds. In this work, we continue to develop and analyze the framework in [13] for the canonical case of real-valued scores R that exhibit *non-decreasing conditional event probabilities*.

Definition 3. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. We say that R has **non-decreasing conditional event probabilities** with respect to $Y \in \{0, 1\}$ if and only if the conditional event probability $\mathbf{P}[Y = 1 \mid R = t]$ is non-decreasing as a function of $t \in \mathbb{R}$. We say that R has **non-decreasing conditional event probabilities** with respect to Y **conditional on A** if for each $a \in \mathcal{A}$, conditional on $A = a$, R has non-decreasing conditional event probabilities with respect to Y .

Having non-decreasing conditional event probabilities is a natural condition and arguably a necessary condition for any reasonable scoring variable. Consider again the example of COMPAS scores. The conditional event probability $\mathbf{P}[Y = 1 \mid R = t]$ is the likelihood that a criminal will re-offend, given an individual's COMPAS Score is t . If COMPAS scores did not exhibit non-decreasing conditional event probabilities then they are assigning a higher score to somebody who's less likely to re-offend than another person with a lower score. This is the opposite of the stated purpose of COMPAS scores.

For another example, consider SAT scores. SAT scores are used by college admissions to assess an applicant's aptitude. Let $Y \in \{0, 1\}$ indicate whether or not a student does well in college by some measure (e.g., whether or not they graduate with at least some GPA). Here, the conditional event probability $\mathbf{P}[Y = 1 \mid \text{SAT} = t]$ is the likelihood that a student will do well, given that their SAT score is t . If the SAT scores did not satisfy non-decreasing conditional event probabilities, then at some score t a higher SAT score would lower our expectations of a student to perform well, which would force us to question the validity and reliability of SAT scores.

A third example, coming from a more theoretical perspective, is the *Bayes optimal regressor* R^* of Y on (X, A) . The Bayes optimal regression is defined as the function $R(X, A)$ minimizing $\mathbf{E}[(Y - R(X, A))^2]$. Thus we can interpret R^* as an *ideal* scoring system for (Y, X, A) . It is not difficult to show that a Bayes optimal regressor R^* satisfies non-decreasing conditional event probabilities on Y , conditional on A . Indeed, for any fixed $a \in \mathcal{A}$ and $t \in \mathbb{R}$, we have

$$\begin{aligned} \mathbf{P}[Y = 1 \mid R^* = t, A = a] &= \mathbf{E}[Y \mid R^* = t, A = a] \\ &= \mathbf{E}[Y \mid R^* = t] = t, \end{aligned}$$

which is an increasing function of t .

There is an extensive literature concerning the study and use of variables that have non-negative conditional event probabilities with respect to some target event³. For example, such variables arise naturally in signal detection theory [8] and in the study of medical diagnoses based on tests [20, 21]. In the medical literature, non-decreasing conditional event probabilities with respect to a medical condition are a desirable property of any medical test used as a basis for diagnosis [20]. There are algorithms towards transforming variables to exhibit non-decreasing conditional event probabilities with respect to some target event, or to adjust noisy empirical data to conform to a hypothesis that the underlying variable has non-decreasing conditional event probabilities [9, 10, 17, 21]. We refer to [12] and the above references for further discussion.

³Some of these discussions are framed in terms of R having non-decreasing likelihood ratio, which is easily shown to be equivalent to non-decreasing conditional event probabilities (e.g., [12]). Other discussions are framed in terms *receiving operator characteristic (ROC) curve* of R . ROC curves are defined in Definition 5 and the connection to non-decreasing conditional event probabilities is given in Lemma 2.1.

1.3.1 Randomized one threshold classifiers

In this work, we are particularly interested in derived predictors $\tilde{Y}(R, A)$ that are randomized k -threshold predictors for $k = 1$; i.e., just one randomized threshold. Randomized one threshold predictors are interesting because they are very easy to understand and interpret on an individual basis. To interpret the action of a randomized one-threshold classifier $\tilde{Y}(R, a)$, fix a demographic $a \in \mathcal{A}$. By definition, there is a threshold $t_a \in \mathbb{R}$ and a probability $p_a \in [0, 1]$ associated with this demographic. Given an individual with score R from demographic a , we look at the individual's score with probability p_a . If the score is greater than or equal to t_a then we return $\tilde{Y}(R, a) = 1$, and if the score is less than t_a then we return $\tilde{Y}(R, a) = 0$. Otherwise, with remaining probability $1 - p_a$, we automatically return $\tilde{Y}(R, a) = 0$. Note all individuals above threshold t_a are treated identically and all individuals below threshold t_a are treated identically. The simplified mechanics of a randomized one-threshold classifier allows for individuals, specifically policy-makers, to directly and subjectively assess the fairness of the mechanism as well as more easily compare the mechanism across demographics.

Compared to a randomized two-threshold classifier, randomized one-threshold classifiers are less expressive. One might even believe that having only one threshold, and treating individuals identically on each side of those thresholds, is too restrictive to satisfy useful notions of equality. Consider the action of a randomized one-threshold classifier in the example of COMPAS scores. For a given individual, with some probability $1 - p_a$, a randomized one-threshold classifier would not consider their COMPAS score or any other data associated with them, and automatically label them as non-recidivist.

Conversely, within a particular notion of equality, if a randomized one-threshold classifier was shown to be optimal in some universal sense, then this would be an interesting structural fact about that notion of equality. Understanding the structure of optimal mechanisms for a fixed notion of equality helps to better understand that notion of equality.

1.3.2 The main theorem

Surprisingly, we show that for the canonical case of scores R that exhibit non-decreasing conditional event probabilities, the optimal derived classifier $\tilde{Y}(R, A)$ satisfying equalized odds can always be obtained within the restricted class of randomized one-threshold classifiers.

Theorem 4. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let ℓ be a loss function assigning nonnegative costs to the false positive and false negative error rates of a predictor $\tilde{Y}(R, A)$.*

The loss-minimizing derived classifier $\tilde{Y}(R, A)$ subject to satisfying equalized odds can always be obtained by a randomized one-threshold classifier. Moreover, an optimal randomized one-threshold classifier subject to equalized odds can be computed efficiently.

We interpret Theorem 4 as an extension of the results in [13], which show that for any R , including those that don't have non-decreasing conditional probabilities, one can compute randomized two-threshold classifiers that are optimal among randomized threshold classifiers. We highlight three key differences. First, Theorem 4 applies only to distributions over (Y, A, R) where R has non-decreasing conditional event probabilities. Second, Theorem 4 produces randomized one-threshold classifiers, a very restrictive subclass of randomized two-threshold classifiers. Third, Theorem 4 is *universal* in the sense that it guarantees an optimal derived classifier over *all* derived classifiers, without restriction on its structure. The second and third points have interesting structural implications that we discuss at length in Section 6. The proofs of the second and third points make essential use of the structural properties of non-decreasing conditional event probabilities.

Theorem 4 and its implications are discussed further in Section 6.

1.4 Organization

The remainder of this article is organized as follows. We give a high level overview of the proof of the Theorem 4 in Section 2. This section identifies two key lemmas, the *Containment Lemma* and the *Constructibility Lemma*. The Containment Lemma is proven in Section 3 and the Constructibility Lemma is proven in Section 4. Theorem 4 is algorithmic, and in Section 5, we implement and compare the randomized one-threshold classifiers produced by Theorem 4 to the algorithm of [13] that produces randomized 2-threshold classifiers. In Section 6, we further interpret and explore the implications of our results, and conclude.

2 High-level proof of the main theorem

In this section we give a high level proof of Theorem 4. First we introduce the notions of ROC curves and hulls. Continuing the approach of [13], these objects play important roles in the proof of Theorem 4. These definitions are followed by structural lemmas about the geometry of ROC curves and hulls of *non-decreasing conditional event probabilities*, in particular. We then introduce the two key lemmas for Theorem 4, called the *Containment Lemma* and the *Constructibility Lemma*. Then, before proving these lemmas, we use them to prove Theorem 4. The Containment Lemma and Constructibility Lemma are proven later in subsequent sections.

2.1 ROC curves and hulls

We first define ROC curves.

Definition 5. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. The **receiving operating characteristic (ROC) curve of R and Y , conditional on a fixed demographic $a \in \mathcal{A}$** , is the function $C_a : \mathbb{R} \rightarrow [0, 1] \times [0, 1]$ defined by

$$C_a(t) = (\mathbf{P}[R > t \mid A = a, Y = 0], \mathbf{P}[R > t \mid A = a, Y = 1]).$$

Note that the ROC curves $C_a(t)$ over $a \in \mathcal{A}$ capture the entire joint distribution (Y, A, R) . ROC curves can also be understood in terms of deterministic classifiers (see Definition 1). Let $a \in \mathcal{A}$ be a fixed demographic, and let $t \in \mathbb{R}$ be a fixed threshold value. Let FP_{t_a} denote the false positive rate, conditional on $A = a$, of a deterministic one-threshold classifier with threshold $t_a = t$ for demographic a . Similarly, let TP_{t_a} denote the corresponding true positive rate. Then,

$$C_a(t) = (\text{FP}_{t_a}, \text{TP}_{t_a})$$

for all t .

We analyze the ROC curves $C_a(t)$ (where $a \in \mathcal{A}$) not only as a function of t , but in terms of the graph it draws in $\mathbb{R}^2$ and its geometric properties. We are particularly interested in the convex hull of these points.

Definition 6. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. For each $a \in \mathcal{A}$, the **ROC hull**, denoted D_a , is defined as the convex hull of the graph of the ROC curve $C_a(t)$,

$$D_a \stackrel{\text{def}}{=} \text{conv}(\{C_a(t) : t \in \mathbb{R}\}),$$

where $\text{conv}(X)$ denotes the convex hull of a set of points X .

2.2 ROC curves and hulls for non-decreasing conditional event probabilities

We require some structural observations about the geometry of ROC curves and hulls for the particular case of non-decreasing conditional event probabilities. The following lemma is well-known (see, e.g., [12]).

Lemma 2.1. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Then the following are equivalent.*

1. *For each $a \in \mathcal{A}$, R has non-decreasing conditional event probabilities with respect to Y conditional on $A = a$.*
2. *For each $a \in \mathcal{A}$, the ROC curve $C_a(t)$ is concave.*

When R has non-decreasing conditional event probabilities, the geometric implications for the ROC curve $C_a(t)$ immediately implies the following facts about the geometry of the ROC hulls D_a , for each $a \in \mathcal{A}$.

Lemma 2.2. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . For each $a \in \mathcal{A}$, the upper envelope of D_a coincides with the ROC curve $C_a(t)$, and the lower envelope coincides with the line segment from $(0, 0)$ to $(1, 1)$.*

2.3 The Containment Lemma

Let us recall, at a high level, the main claims of Theorem 4. Theorem 4 asserts that the optimal derived classifier $Y^*(R, A)$, subject to equalized odds, can always be obtained by one randomized threshold. Here optimality is measured by a loss function that assigns costs for false positive and false negative errors. *A priori*, since we assume no structure of $Y^*(R, A)$ beyond the constraints under which it is optimal (namely, that Y^* satisfies equalized odds and is derived from R and A), it would seem difficult to limit the possible false positive and false negative rates that it might achieve.

The first key lemma, which we call the *Containment Lemma*, sets out to limit of the set of values that the false positive and false negative rates of an optimal classifier could take. In particular, the Containment Lemma asserts that the false positive and true positive rates, as coordinates, are contained in the ROC hull D_a for all demographics $a \in \mathcal{A}$ simultaneously.

Lemma 2.3. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $Y^*(R, A)$ be a optimal classifier derived from R , conditional on A , subject to equalized odds, with respect to some nonnegative cost function of the false positive and false negative error rates. Let FP^* be the false positive rate of Y^* and let TP^* be the true positive rate of Y^* . Then $(\text{FP}^*, \text{TP}^*) \in \bigcap_{a \in \mathcal{A}} D_a$.*

2.4 The Constructibility Lemma

The Containment Lemma, suprisingly, sets limits to the false positive and true positive rates that the optimal predictor Y^* can obtain. In particular, the optimum false positive and true positive rates are contained in the intersection $\bigcap_{a \in \mathcal{A}} D_a$. Recall that Theorem 4 asserts that the optimal predictor can be matched by one randomized threshold. Given the Containment Lemma, it suffices to show that any false positive and true positive rate in $\bigcap_{a \in \mathcal{A}} D_a$ can be achieved by a single randomized threshold. This is the claim of the following lemma, which we call the *Constructibility Lemma* since it is where we actually construct the randomized one-threshold classifier.

Lemma 2.4. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $(x, y) \in \bigcap_{a \in \mathcal{A}} D_a$. Then there exists a randomized one-threshold classifier $\tilde{Y}(R, A)$ with the following properties.*

1. $\tilde{Y}$ satisfies equalized odds with respect to A .
2. For each $a \in \mathcal{A}$, conditional on $A = a$, $\tilde{Y}(R, A)$ has false positive rate x and true positive rate y .

Moreover, this randomized one-threshold classifier can be computed efficiently.

We point out that [13] previously showed that any point in $\bigcap_{a \in \mathcal{A}} D_a$ can be obtained by a randomized two threshold classifier, without assuming non-increasing conditional event probabilities; see also Remark 4.1.

2.5 Proving the main theorem

Let us prove the main theorem, Theorem 4, assuming for the time being that Lemma 2.3 and Lemma 2.4 hold true. We first restate Theorem 4 for the reader's convenience.

Theorem 4. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let ℓ be a loss function assigning nonnegative costs to the false positive and false negative error rates of a predictor $\tilde{Y}(R, A)$.*

The loss-minimizing derived classifier $\tilde{Y}(R, A)$ subject to satisfying equalized odds can always be obtained by a randomized one-threshold classifier. Moreover, an optimal randomized one-threshold classifier subject to equalized odds can be computed efficiently.

Proof. Let $Y^*(R, A)$ be an optimal derived predictor subject to equalized odds. Let FP^* be the false positive rate of Y^* , and let TP^* be the true positive rate of Y^* . Because Y^* satisfies equalized odds, these are also the false positive and true positive rates conditional on $A = a$ for all $a \in \mathcal{A}$. By Lemma 2.3, because R has non-decreasing conditional event probabilities and Y^* is an optimal derived predictor, $(\text{FP}^*, \text{TP}^*)$ is in all of the ROC hulls; i.e., $(\text{FP}^*, \text{TP}^*) \in \bigcap_{a \in \mathcal{A}} D_a$.

By assumption, the loss function is a linear function of FP and TP . To identify FP^* and TP^* , it suffices to minimize the loss function over $\bigcap_{a \in \mathcal{A}} D_a$. Since the loss is increasing in FP and decreasing in TP , it suffices to search the upper envelope of $\bigcap_{a \in \mathcal{A}} D_a$. By Lemma 2.2, the upper envelope of $\bigcap_{a \in \mathcal{A}} D_a$ coincides by the pointwise minima (in $\mathbb{R}^2$) over the ROC curves, and we can search for $(\text{FP}^*, \text{TP}^*)$ efficiently by ternary search (similar to [13], except here slightly simpler).⁴

Suppose now that we have the optimum false positive and true positive rates $(\text{FP}^*, \text{TP}^*)$. By Lemma 2.4, because (a) R has non-decreasing conditional event probabilities with respect to Y conditional on A , and (b) $(\text{FP}^*, \text{TP}^*) \in \bigcap_{a \in \mathcal{A}} D_a$, we can also obtain the rates $(\text{FP}^*, \text{TP}^*)$ via a randomized single-threshold classifier. This completes the proof. $\blacksquare$

3 Proof of the Containment Lemma

In this section, we prove the Containment Lemma (Lemma 2.3). At a high level, the Containment Lemma asserts that any optimal classifier satisfying equalized odds and derived from R and A has false positive

⁴Alternatively, because the objective is a linear function and each D_a is convex with upper envelope explicitly given by the ROC curve $C_a(t)$, one can apply the ellipsoid algorithm, where we separate over $\bigcap_{a \in \mathcal{A}} D_a$ by querying appropriate points of the ROC curves, $C_a(t)$.

and true positive coordinates contained in the intersection of ROC hulls, $\bigcap_{a \in \mathcal{A}} D_a$. This is a necessary step towards showing that randomized threshold classifiers can achieve optimality (let alone with just one randomized threshold).

3.1 Monotonic classifiers

Definition 7. Let (A, R) be jointly distributed random variables with $A \in \mathcal{A}$ for a finite set $\mathcal{A}$ and $R \in \mathbb{R}$. A derived predictor $\tilde{Y}(R, A) \in \{0, 1\}$ is **monotonically non-decreasing conditional on A** , or simply **monotone conditional on A** , if for all $a \in \mathcal{A}$,

$$\mathbf{P}[\tilde{Y}(R, A) = 1 \mid A = a, R = t]$$

is monotonically non-decreasing in t almost everywhere.

Alternatively, $\tilde{Y}$ is monotonically non-decreasing conditional on A if R has non-decreasing conditional event probabilities with respect to $\tilde{Y}(R, A)$, conditional on A .

Monotonic classifiers are “natural” for the setting where R has non-decreasing conditional event probabilities, in the following sense. By definition, an increase in R can only increase the likelihood of the event we are trying to predict. Any “good” classifier, then, *ought* to be monotonic. However intuitive, this argument does not directly imply that optimal classifiers subject to equalized odds are monotonic, since these classifiers may break monotonicity in order to satisfy the equalized odds constraints.

3.2 Containing monotone classifiers

Let us first prove an analogue Lemma 2.3 for the special case of monotonic classifiers. Note that unlike Lemma 2.3, the following lemma does not require the derived predictor $\tilde{Y}(R, A)$ to satisfy equalized odds, or require it to be optimal in any way.

Lemma 3.1. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $\tilde{Y}(R, A)$ be a non-decreasing classifier derived from R , conditional on A . For $a \in \mathcal{A}$, let FP_a be the false positive rate of $\tilde{Y}(R, A)$ conditional on $A = a$, and let TP_a be the true positive rate of $\tilde{Y}(R, A)$ conditional on $A = a$. For all $a \in \mathcal{A}$, we have

$$(\text{FP}_a, \text{TP}_a) \in D_a.$$

Proof. Fix $a \in \mathcal{A}$. Suppose first that $\mathbf{E}[\tilde{Y}(R, a)]$ is a monotonically increasing step function with finitely many steps. Then $\tilde{Y}(R, a)$ can be written as a convex combination of a finite number of single-threshold functions. In turn, the false positive and true positive rates $(\text{FP}_a, \text{TP}_a)$ is the same convex combination of the points corresponding to the thresholds on the ROC curve $C_a(t)$. By definition of D_a as the convex hull of the ROC curve $C_a(t)$, we have $(\text{FP}_a, \text{TP}_a) \in D_a$. In the general setting, we observe that D_a is closed and convex, and take $\tilde{Y}(R, a)$ as the limit of convex combinations of single-threshold functions. As the limit of a sequence of points in D_a , we have $(\text{FP}_a, \text{TP}_a) \in D_a$. ■

3.3 Dominating non-monotonic classifiers

The following definitions imposes a partial order on the family of predictors.

Definition 8. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let $Y'(R, A)$ and $Y''(R, A)$ be two classifiers derived from (R, A) . For $a \in \mathcal{A}$, let FP'_a and FP''_a and be the false positive rates of $Y'(R, A)$ and $Y''(R, A)$, respectively, conditional of $A = a$. Let FN'_a and FN''_a and be the false negative rates of $Y'(R, A)$ and $Y''(R, A)$, respectively, conditional of $A = a$.

We say that $Y'(R, A)$ **dominates** $Y''(R, A)$ **conditional on A** if for all $a \in \mathcal{A}$, conditional on $A = a$, $Y'(R, A)$ has false positive and false negative rates less than or equal to that of $Y''(R, A)$. That is, if

$$FP'_a \leq FP''_a \text{ and } FN'_a \leq FN''_a.$$

We say that $Y'(R, A)$ **strictly dominates** $Y''(R, A)$ if not all conditional false positive and false negative rates are equal. For $\epsilon > 0$, we say that $Y'(R, A)$ **ϵ -dominates** $\tilde{Y}(B, A)$ if for all $a \in \mathcal{A}$,

$$FP'_a \leq FP''_a + \epsilon \text{ and } FN'_a \leq FN''_a + \epsilon.$$

Observe that if Y' dominates Y'' , then it will have a better objective value with respect to *any* cost function of false positive and false negative rates.

The following lemma formalizes the informal suggestion, from the discussion after Definition 7, that classifiers can only benefit by being monotonic.

Lemma 3.2. Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $Y'(R, A)$ be a classifier derived from R , conditional on A . For all $\epsilon > 0$, there exists a monotonic classifier $Y''(R, A)$ that ϵ -dominates $\tilde{Y}(R, A)$ conditional on A .

Proof. Let $Y'(R, A)$ be a classifier that is not monotonic. Suppose first that for each $a \in \mathcal{A}$, conditional on $A = a$, $Y'(R, A)$ is a step function defined over a finite number of intervals. Fix $a \in \mathcal{A}$. Conditional on $A = a$, let $Y'(R, A)$ be a step function defined over k consecutive intervals $I_1, \dots, I_k$. That is, for each interval I_i , there is a fixed probability p_i such that for all $r \in I_i$, $p_i = \mathbf{P}[Y'(R, A) | R = r, A = a]$. For each i , let $r_i = \mathbf{P}[R \in I_i | A = a]$. Dropping any intervals with probability 0, we assume $r_i > 0$ for all i . If $Y'(R, A)$ is not monotone conditional on $A = a$, then there are consecutive intervals I_i and I_{i+1} such that $p_i > p_{i+1}$. Let $J = I_i \cup I_{i+1}$. Consider the predictor obtained from Y' as follows. Outside of $A = a$ and $R \in J$, return Y' . For $A = a$ and $R \in J$, label 1 with probability $(r_i p_i + r_{i+1} p_{i+1}) / (r_i + r_{i+1})$ and 0 otherwise. It is straightforward to verify that, because R has non-decreasing conditional event probabilities, the new predictor dominates the previous one⁵. We have also decreased the number of intervals in the step function by 1, and can do so only a limited number of times before the obtained predictor is monotone (conditional on $A = a$). By applying such replacements to all $a \in \mathcal{A}$, we obtain a predictor $Y''(R, A)$ that dominates $Y'(R, A)$ and is monotone.

We have shown that if $\tilde{Y}(R, A)$ is a finite step function conditional $A = a$ for each $a \in \mathcal{A}$, it is dominated by some monotone predictor. In the general case, for each $a \in \mathcal{A}$, we approximate $\tilde{Y}(R, A)$ conditional on $A = a$ by a finite step function to arbitrary precision. In particular, we obtain a predictor $Y''(R, A)$ where:

1. For all $a \in \mathcal{A}$, conditional on $A = a$, $Y''(R, A)$ is a step function.
2. For all $a \in \mathcal{A}$, the false positive and true positive rates of $Y''(R, A)$ conditional on $A = a$ are within an ϵ -additive factor of the false positive and true positive rates of $\tilde{Y}(R, A)$.

The second property implies that $Y''(R, A)$ ϵ -dominates $\tilde{Y}(R, A)$. The first property implies there exists a monotone classifier $Y''(R, A)$ that dominates $Y'(R, A)$, and in turn $Y''(R, A)$ ϵ -dominates $\tilde{Y}(R, A)$, as desired. ■

⁵Informally, the intuition is that we are shifting probability from the lower interval I_i to the higher interval I_{i+1} , where by assumption on R , the probability is put to better use.

3.4 Proving the Containment Lemma

Finally, we prove the Containment Lemma. In light of Lemma 3.1, the remaining effort is to address the case of predictors satisfying equalized odds that are not *a priori* monotone. Again, we point out that despite Lemma 3.2, we do not know that the optimal predictor satisfying equalized odds is monotone or close to monotone *a priori*, because Lemma 3.2 does not preserve equalized odds. We first restate the Containment Lemma for the reader's convenience.

Lemma 2.3. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $Y^*(R, A)$ be a optimal classifier derived from R , conditional on A , subject to equalized odds, with respect to some nonnegative cost function of the false positive and false negative error rates. Let FP^* be the false positive rate of Y^* and let TP^* be the true positive rate of Y^* . Then $(\text{FP}^*, \text{TP}^*) \in \bigcap_{a \in \mathcal{A}} D_a$.*

Proof. By Lemma 2.2, it suffices to show that the coordinates $(\text{FP}^*, \text{TP}^*)$ lie on or under the ROC curve $C_a(t)$ for all $a \in \mathcal{A}$, and on or above the line segment from $(0, 0)$ to $(1, 1)$.

We first address the second point. We first observe that for any $t \in [0, 1]$, the false positive and true positive coordinates (t, t) is obtained by an oblivious randomized that on any given instance, labels the instance 1 with probability t and 0 with probability $1 - t$. Clearly, such a predictor satisfies equalized odds as well. Since Y^* is optimal, it cannot be strictly dominated by any of these predictors. Thus $(\text{FP}^*, \text{TP}^*)$ lies on or above the line segment from $(0, 0)$ to $(1, 1)$.

It remains to show that the coordinates $(\text{FP}^*, \text{TP}^*)$ lie on or under the ROC curve $C_a(t)$ for all a . By Lemma 3.2, for all $\epsilon > 0$, there exists a monotone derived predictor $\tilde{Y}(R, A)$ that ϵ -dominates Y^* . For each $a \in \mathcal{A}$, let FP_a be the false positive rate of $\tilde{Y}(R, A)$ conditional on $A = a$, and let TP_a be the true positive rate of $\tilde{Y}(R, A)$ conditional on $A = a$. We have $\text{TP}^* \leq \text{TP}_a + \epsilon$ and $\text{FP}^* \geq \text{FP}_a - \epsilon$ for all $a \in \mathcal{A}$ because $\tilde{Y}(R, A)$ ϵ -dominates $Y^*(R, A)$. Moreover, by Lemma 3.1, for all $a \in \mathcal{A}$, we have $(\text{FP}_a, \text{TP}_a) \in D_a$, and in particular, $(\text{FP}_a, \text{TP}_a)$ lies under the curve $C_a(t)$. Given the relative position $(\text{FP}^*, \text{TP}^*)$ to $(\text{FP}_a, \text{TP}_a)$, we see that for all $\epsilon > 0$ and all $a \in \mathcal{A}$, the point $(\max\{\text{FP}^* + \epsilon, 1\}, \min\{\text{TP}^* - \epsilon, 0\})$ lies under the curve $C_a(t)$ as well. Since the area on or under the ROC curve $C_a(t)$ intersected with the unit square $[0, 1] \times [0, 1]$ is closed and bounded, taking the limit as $\epsilon \rightarrow 0$, we conclude that $(\text{FP}^*, \text{TP}^*)$ lies on or under the curve $C_a(t)$ for all $a \in \mathcal{A}$. ■

4 Proof of the Constructibility Lemma

In this section, we prove the second key lemma supporting Theorem 4, the Constructibility Lemma. Recall that the high level claim in Theorem 4 is that we can derive a randomized one-threshold classifier satisfying equalized odds competitive with *any* derived classifier satisfying odds. Previously, the Containment Lemma (Lemma 2.3) showed that any optimal derived classifier satisfying equalized odds has its false positive and true positive error rates contained in the intersection $\bigcap_{a \in \mathcal{A}} D_a$. It remains to show that any such point in $\bigcap_{a \in \mathcal{A}} D_a$ can also be achieved by a randomized single threshold classifier, and that this classifier can be constructed efficiently. This second step is given by the Constructibility Lemma. We first restate the Constructibility Lemma for the reader's convenience.

Lemma 2.4. *Let (Y, A, R) be jointly distributed random variables with $Y \in \{0, 1\}$, $A \in \mathcal{A}$ for a finite set $\mathcal{A}$, and $R \in \mathbb{R}$. Let R have non-negative conditional event probabilities with respect to Y , conditional on A . Let $(x, y) \in \bigcap_{a \in \mathcal{A}} D_a$. Then there exists a randomized one-threshold classifier $\tilde{Y}(R, A)$ with the following properties.*

1. $\tilde{Y}$ satisfies equalized odds with respect to A .

2. For each $a \in \mathcal{A}$, conditional on $A = a$, $\tilde{Y}(R, A)$ has false positive rate x and true positive rate y .

Moreover, this randomized one-threshold classifier can be computed efficiently.

Proof. For each $a \in \mathcal{A}$, we will choose $t_a \in \mathbb{R}$ and $p_a \in [0, 1]$ such that conditional on $A = a$, the randomized one-threshold classifier with respect to these probabilities and thresholds (per Definition 2) has false positive rate x and true positive rate y . This would establish the second claim in the lemma. Once the second claim is established, the first claim is immediate. Indeed, by construction, the obtained randomized one-threshold classifier necessarily has the same false positive rates, conditional $A = a$, across $a \in \mathcal{A}$, and likewise for its true positive rates.

Fix $a \in \mathcal{A}$. By Lemma 2.2, (x, y) lies under the curve C_a . Consider the line segment starting from $(0, 0)$ and passing through (x, y) , continuing until it intersects with the curve C_a at a point $C(t_a)$ for some t_a . Since (x, y) is on the line segment between $(0, 0)$ and $C_a(t)$, we can write (x, y) as a convex combination of $(0, 0)$ and $C_a(t)$ as a convex combination

$$(x, y) = p_a C_a(t) + (1 - p_a)(0, 0) = p_a C_a(t).$$

To produce a randomized one-threshold classifier, we interpret $(0, 0)$ as the false positive rate and true positive rate of the constant classifier that labels all instances 0. Then (x, y) can be obtained by the randomized one-threshold classifier that with probability p_a , applies the single-threshold classifier I_{t_a} , and otherwise answers 0.

Lastly, we observe that the above proof of is constructive, and suggests a simple algorithm for constructing the desired randomized one-threshold classifier. For fixed $a \in \mathcal{A}$, to find the desired threshold t_a and p_a , one can binary search for t_a within the range of R . Given a query point t , one can consider the line from $(0, 0)$ to $C_a(t)$ and decide if it passes above, below, or through the target (x, y) . This decision dictates whether to move right, move left, or stop in the binary search. ■

Remark 4.1. [13] proved that for any real-valued score R , with or without non-decreasing conditional event probabilities, the point (x, y) can be obtained by a randomized two-threshold classifier. The proof is also based on the geometry of convex hulls in $\mathbb{R}^2$. The main technical difference here is that when R has non-decreasing conditional event probabilities, for each $a \in \mathcal{A}$, the upper envelope of the ROC hull D_a coincides with the ROC curve $C_a(t)$ (Lemma 2.2). This nicer geometric structure for non-decreasing conditional event probabilities ultimately yields randomized one-threshold classifiers.

5 Empirical results

In this section, we describe the empirical results of experiments to supplement and help validate the theoretical discussion above. One of the motivations of these empirical investigations is that the theoretical conclusions above are predicated on the scoring variable R to have non-decreasing conditional event probabilities, conditional on each $a \in \mathcal{A}$. While we expect any reasonable, real world scoring variable to *aspire*, for the various reasons given in Section 1.3, real world scoring variables may not satisfy non-decreasing conditional probabilities perfectly (i.e., for all t). Moreover, the empirical data produced by real world scoring functions may be noisy that creates further deviations. A partial goal of the empirical experiments is to understand the extent to which these real-world deviations from the theoretical framework impact the overall conclusion.

To this end, we compute and compare the randomized one-threshold predictors produced by Theorem 4 with the randomized two-threshold predictors produced by the algorithm in [13]. We test these algorithms on datasets of both FICO scores and COMPAS scores (each described in greater detail below, in their respective function). We were particularly interested in understanding, over a wide range of loss functions, whether the loss value was similar for both the algorithms. Within that, we were also interested in understanding if the

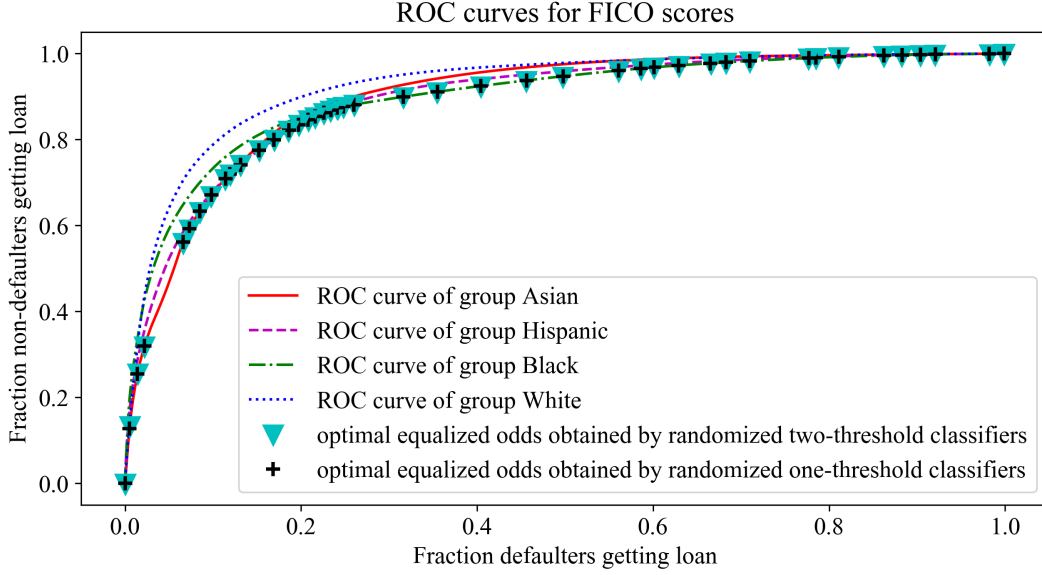


Figure 1: Optimal false positive and true positive rates of randomized one threshold and two threshold predictors derived from FICO scores satisfying equalized odds for different cost ratios. Observe that the optimal false positive and true positive rates for both these predictors are closely equal for any cost ratio.

false positive and true positive error rates obtained by the two algorithms (for a particular loss function) are generally similar or tend to differ. We also paid attention to the actual probabilities p_a and thresholds t_a , for $a \in \mathcal{A}$, in the optimal randomized one-threshold classifier. In particular, we wanted to understand if these classifiers tend to be very randomized (if p_a close to $1/2$) or more deterministic (with p_a close to 0 or 1), and whether these classifiers tend to use the scoring variable (with p_a close to 1) or tend to be oblivious to it (with p_a close to 0.)

In particular, we ran the algorithm proposed by [13], which produces a randomized two-threshold classifier, and the algorithm from Theorem 4, which assumes non-decreasing conditional event probabilities for a score and produces a randomized one-threshold predictor, on FICO and COMPAS dataset. We show that, for any loss function ℓ , the false positive rate FP and true positive rate TP of the two classifiers are approximately (negligible difference) equivalent. Additionally, we obtain parameters, $t_a \in \mathbb{R}$ and $p_a \in [0, 1]$ of the optimal randomized one-threshold classifier, derived from FICO or COMPAS scores, for each $a \in \mathcal{A}$. These parameters can be found efficiently with a binary search.

We concretely describe the loss function of each instance in the experiments. For a fixed loss function, we have a non-negative cost $c_{FP} \geq 0$ for false positive error, and a non-negative cost $c_{FN} \geq 0$ for false negative error. Given a classifier $\tilde{Y} \in \{0, 1\}$ trying to predict $Y \in \{0, 1\}$, the loss of $\tilde{Y}$ is given by

$$\begin{aligned} \ell(\tilde{Y}) &= \mathbf{E}[c_{FP} 1_{\tilde{Y}=1, Y=0} + c_{FN} 1_{\tilde{Y}=0, Y=1}] \\ &= c_{FP} \mathbf{P}[\tilde{Y} = 1, Y = 0] + c_{FN} \mathbf{P}[\tilde{Y} = 0, Y = 1]. \end{aligned}$$

Here $1_{\tilde{Y}=1, Y=0}$ is the $\{1, 0\}$ -indicator of the event of a true positive error, and $1_{\tilde{Y}=0, Y=1}$ is the $\{1, 0\}$ -indicator of the event of a false negative error.

Ultimately, the optimization problem is scale invariant, and the optimality is determined by the proportions of c_{FP} to c_{FN} . The **cost ratio** of a loss function parametrized by c_{FP} and c_{FN} is given by

$$\frac{c_{FP}}{c_{FP} + c_{FN}}.$$

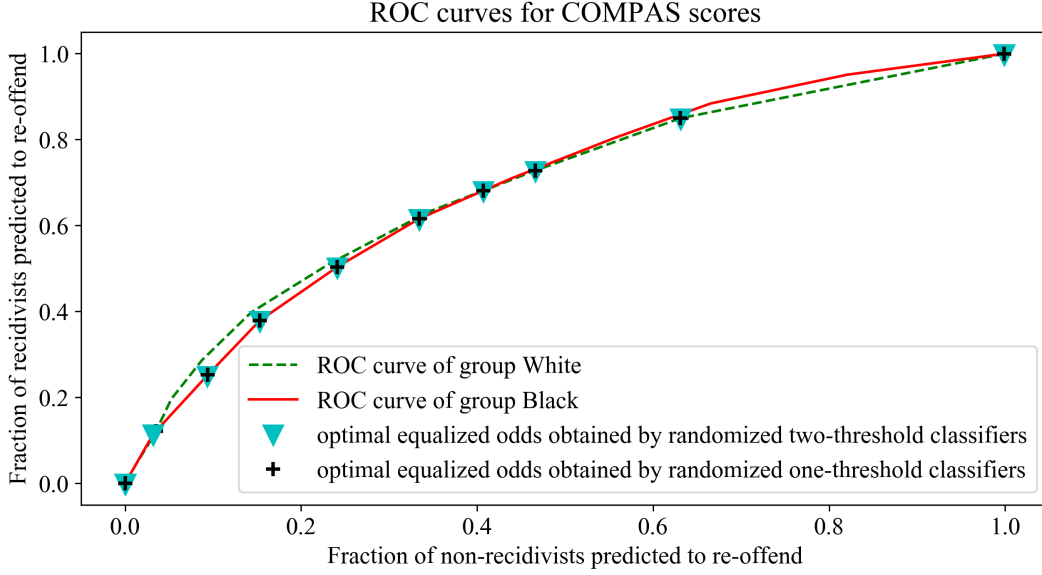


Figure 2: Optimal false positive and true positive rates of randomized one threshold and two threshold predictors derived from COMPAS scores satisfying equalized odds for different cost ratios. Observe that the optimal false positive and true positive rates for both these predictors are closely equal for any cost ratio.

We ran the two algorithms for a range of cost ratios from 0 to 1.

For the wide range of cost ratios from 0 to 1, we generally found that the optimal randomized one-threshold classifiers (produced by Theorem 4) and that the optimal randomized two-threshold classifiers (per [13]) are very similar. The optimum cost in either category tended to be similar, with the optimal randomized one-threshold classifiers often that matching the optimal randomized two-threshold classifiers. Moreover, the obtained false positive and true positive coordinates were very similar and oftentimes the same. Perhaps this is not too surprising considering that the ROC curves for both these data sets are mostly concave. We also observed that the randomized one-threshold classifiers are nearly deterministic (with probabilities p_a equal or close to 1) when the cost ratio is fairly low, and more randomized when the cost ratio is high. This is partly because the “default label” for randomized thresholds was defined to be 0. If we had instead defined it to be 1, then we would expect more deterministic behavior for high cost ratio and more randomized behavior for low cost ratios. In practice, one probably prefers to choose, based on the target cost ratio, the orientation that generates a more deterministic classifier. Lastly, we observe that the randomized one-threshold classifiers generally produced probabilities greater than .5, and therefore are not oblivious to the scoring function R . Similar to the previous discussion on determinism, low cost ratios (“oriented” with the default label 0) produced less oblivious classifiers.

5.1 FICO scores

As mentioned earlier, FICO scores are used in the United States to make loan decisions. These scores predict credit worthiness. The FICO data is based on TransUnion Transrisk scores from 2003 [22]. These scores were rescaled in the dataset to the range from 0 to 100. We have four demographics based on the ethnicity of the loan applicant. Classifiers derived from FICO scores predict whether an individual with a FICO score t will fail to payback (or default) their loan. The true outcome Y of individuals is based on whether, after an 18-24 month period, they succeed or fail to pay back loans on all their accounts within 90 days. A positive outcome

($Y = 1$) indicates that an individual did not default any loan. We ran our algorithms from data extracted from the library [14], which are derived from the FICO scores on a dataset based on TransUnion Transrisk scores [22]. We ran both algorithms for all cost ratios from 0 to 1, in increments of .02.

In Figure 1, we plot the loss minimizing false positive rate and true positive rate obtained from by the optimal randomized one-threshold classifier and the optimal randomized two-threshold classifier. The data points from the two algorithms for the same cost ratio are generally overlayed on top of one another, as can be confirmed in the supplementary tables. Table 1 in the supplement lists the outputs (namely the true positive rates, the false positive rates, and the ratio of losses) of the two algorithms, for each cost ratio from 0 to 1 in increments of .02. These values all tend to be very similar between the two classifiers. Table 2 in the supplement gives the thresholds t_a and probabilities p_a for each demographic in the optimal randomized one-threshold classifier. The probabilities p_a were 1 or close to 1 for low and moderate cost ratios and tended roughly to .5 as the cost ratio approached 1, only dropping below .5 for one nontrivial probability. These data points for FICO scores are strongly aligned with the general findings described above.

5.2 COMPAS scores

We also ran the two algorithms on COMPAS scores. COMPAS scores, ranging from 1 to 10, are used to predict criminal recidivism. We applied to the algorithms to the COMPAS data used and provided by Angwin et al. [1]. The data obtained by Angwin et al. [1] contains two years of COMPAS scores from the Broward County Sheriff’s Office in Florida. They recieved data from 18610 people who were scored in the years 2013 and 2014. Angwin et al. [1] label individuals as recidivists if they had been arrested within two years of obtaining their COMPAS score. This gives us our true outcome Y . There are two demographics corresponding based on the race of the individual. We ran both algorithms for all cost ratios from 0 to 1, in increments of .02.

In Figure 2, we plot the loss minimizing false positive rate and true positive rate obtained from by the optimal randomized one-threshold classifier and the optimal randomized two-threshold classifier. The data points from the two algorithms for the same cost ratio are (for all but one instance) overlayed on top of one another, as can be confirmed in the supplementary tables. Table 2 in the supplement lists the outputs (namely the true positive rates, the false positive rates, and the ratio of losses) of the two algorithms, for each cost ratio from 0 to 1 in increments of .02. These values were essentially identical for all but a few cost ratios. Table 3 in the supplement gives the thresholds t_a and probabilities p_a for each demograaphic in the optimal randomized one-threshold classifier. The probabilities p_a were 1 or close to 1 for low and moderate cost ratios and tended roughly to .5 for one of the demographics as the cost ratio approached 1. Due to the coarseness of COMPAS scores, running the algorithm over the full range of cost ratios produced few distinct false positive and true positive error rates. This can be observed both in the plot in Figure 2 and in the supplementary tables. All in all, we found that the empirical results for COMPAS scores agree with the general findings described above.

6 Discussion and conclusion

In this section, we discuss further implications and interpretations of our results, and summarize our findings in a conclusion.

Universality. Within the important setting of scoring functions with non-decreasing conditional event probabilities, the guarantees of Theorem 4 for finding the optimum classifier subject to equalized odds are universal, producing a classifier provably as good as any possible derived classifiers. By contrast, the [13] results are only proven to be optimal within the search space of threshold functions explicitly considered by the algorithm. Letting aside the fact that the algorithm here is different and the output here has additional nice

properties, this work also shows that the existing algorithm of [13] is universal in the sense described above (for non-decreasing conditional event probabilities). Indeed, we show that optimality can be obtained by a single randomized threshold classifier, and the algorithm of [13] searches over single randomized threshold classifiers as a special case of randomized two threshold classifiers. Previously it was pointed out in [6] that the [13] algorithm is universal for the different fairness criteria of equalized opportunity⁶ where R is specifically the Bayes optimal regressor. We point out that for a qualitatively different model, but similar in spirit and complementing the work here, [6] was able to give optimal classifiers for a different class of objective functions subject to different kinds of equality constraints, using only one deterministic threshold.

Reducing the complexity of optimal classifiers subject to equalized odds. We show that a universally optimal classifier can be obtained, in particular, by a randomized one threshold classifier. We found this surprising because the classifiers from [13] were already very simple. Overall, Theorem 4 establishes an extremely simple and radically transparent type of mechanism (randomized one-threshold classifiers) that can optimally achieve equalized odds. In particular, this makes it more accessible and interpretable. We have observed that one can interpret this mechanism in a few different ways. On one hand, it is clearly valuable to have as concrete an understanding as possible for any algorithmic policy that is actually adopted and has real effect on individuals. Having only one threshold across demographics makes it easier to compare and evaluate (subjectively) the implications for each demographics in real situations. On the other hand, randomized one-threshold classifiers are so crude that, one might argue, it could not capture certain subtle aspects of equality (in the general sense of the word). Even so, Theorem 4 still implies that the model, which is trying to derive an optimal binary classifier from a score R and demographic $a \in \mathcal{A}$ that satisfies equalized odds, does not distinguish between an optimal randomized one-threshold classifier or some other complex classifier that might embody additional qualities to address these concerns. From the focused viewpoint of the model, a randomized one-threshold classifier is at least as good as any other classifier satisfying equalized odds. If a policy maker believes that one randomized threshold is insufficient to achieve fairness (in the general and subjective sense), then this implies that we should look beyond equalized odds to capture other dimensions of fairness.

Applicability to real scoring functions. Our investigations are predicated by an assumption that the score R satisfies non-decreasing conditional event probabilities with respect to the predicted event on a per-demographic basis. We have argued that this is a natural condition and have listed several real-world examples where this property is generally assumed. Of course, real world data, even if sampled from an idealized distribution, may not perfectly conform to these properties. (For example, any finite data is not perfectly smooth.) Since real-world scores are generally *designed* to satisfy non-decreasing conditional event probabilities, we expect the discrepancies of real data from our idealized assumptions to be minimal. In particular, we expect the algorithms designed here for non-decreasing conditional event probabilities to perform well on real data sets.⁷ To validate this hypothesis, we ran the new algorithms, as well as the algorithms of [13], on data sets for COMPAS and FICO scores. These data sets, which are discretized and noisy, do not strictly satisfy non-decreasing conditional event probabilities, but the overall shape of the plotted ROC curves are mostly concave as we would expect. We discussed our findings in greater detail in Section 5 and here we summarize the key takeaways. We found that our algorithm and the algorithm of [13] obtain the same loss value for almost all parametrizations of the objective function, even though the new algorithm here searches only a subset of the space of functions searched by the [13] algorithm. The two algorithms also tend to select very similar FP and TP coordinates. We also inspected the randomized one threshold classifiers

⁶More directly [6] discusses equalized false positive rates, which is similar.

⁷One might also argue that, if the data was (in principle) was supposed to satisfy non-decreasing conditional event probabilities, then running an algorithm (such as ours) specifically for this property can be (informally) understood as a sort of regularization.

produced by the new algorithm, and found that oftentimes the probabilities were close to 1, making these algorithms “almost” deterministic.

Conclusion. We design and analyze derived classifiers subject to equalized fairness for the important setting of scoring variables that satisfy non-decreasing conditional event probabilities (per demographic). Our approach lies within the general framework laid out by [13]. We showed in particular results that strengthen and specialize those of [13], by producing simpler derived classifiers that satisfy a truly universal optimality guarantee. Our assumption of non-decreasing conditional event probabilities was validated by experiments where the specialized algorithm proposed in this work is competitive with the broader algorithm of [13] on real-world data. As a final and more general comment, we point out that – given the emergence of various impossibility theorems that disallow certain one-size-fits-all solutions to algorithmic fairness, and the emphasis on transparency in policy design – identifying and assuming useful and natural structure in the scoring functions – such as non-decreasing conditional event probabilities in our work here – can allow for stronger or cleaner theoretical results that may still be wieldy in modeling real-world phenomena.

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cost ratio	FP_1	FP_2	TP_1	TP_2	ℓ_1/ℓ_2
0.000	0.999	0.999	1.000	1.000	1.000
0.020	0.999	0.999	1.000	1.000	1.000
0.040	0.982	0.982	1.000	1.000	1.000
0.060	0.921	0.921	0.998	0.998	1.000
0.080	0.904	0.904	0.998	0.998	1.000
0.100	0.883	0.883	0.997	0.997	1.000
0.120	0.862	0.862	0.996	0.996	1.000
0.140	0.811	0.811	0.993	0.992	1.000
0.160	0.785	0.785	0.990	0.990	1.000
0.180	0.777	0.777	0.990	0.990	1.000
0.200	0.710	0.710	0.983	0.983	1.000
0.220	0.683	0.683	0.980	0.980	1.000
0.240	0.665	0.666	0.978	0.978	1.000
0.260	0.629	0.629	0.973	0.973	1.000
0.280	0.601	0.601	0.968	0.968	1.000
0.300	0.586	0.586	0.966	0.966	1.000
0.320	0.561	0.561	0.961	0.961	1.000
0.340	0.498	0.498	0.947	0.947	1.000
0.360	0.456	0.456	0.938	0.938	1.000
0.380	0.404	0.404	0.925	0.925	1.000
0.400	0.355	0.354	0.911	0.911	1.000
0.420	0.316	0.316	0.900	0.900	1.000
0.440	0.261	0.260	0.882	0.882	1.000
0.460	0.260	0.260	0.881	0.882	1.000
0.480	0.248	0.248	0.877	0.877	1.000
0.500	0.242	0.242	0.875	0.875	1.000
0.520	0.241	0.241	0.874	0.874	1.000
0.540	0.241	0.241	0.874	0.874	1.000
0.560	0.241	0.241	0.874	0.874	1.000
0.580	0.241	0.241	0.874	0.874	1.000
0.600	0.241	0.241	0.874	0.874	1.000
0.620	0.241	0.241	0.874	0.874	1.000
0.640	0.234	0.233	0.869	0.868	1.000
0.660	0.226	0.226	0.862	0.862	1.000
0.680	0.216	0.216	0.854	0.853	1.000
0.700	0.208	0.208	0.846	0.846	1.000
0.720	0.197	0.198	0.835	0.835	1.000
0.740	0.186	0.186	0.822	0.821	1.000
0.760	0.169	0.169	0.800	0.800	1.000
0.780	0.152	0.152	0.775	0.776	1.000
0.800	0.131	0.131	0.741	0.742	1.000
0.820	0.119	0.119	0.719	0.720	1.000
0.840	0.114	0.114	0.709	0.709	1.000
0.860	0.098	0.097	0.671	0.669	1.000
0.880	0.085	0.085	0.634	0.633	1.000
0.900	0.073	0.073	0.592	0.591	1.000
0.920	0.066	0.066	0.562	0.561	1.000
0.940	0.022	0.022	0.319	0.320	1.000
0.960	0.014	0.014	0.254	0.256	1.000
0.980	0.005	0.005	0.128	0.134	1.000
1.000	0.000	0.000	0.000	0.000	1.000

Table 1: False positive rates, FP_1, FP_2 and true positive rates, TP_1, TP_2 and ratios of the losses ℓ_1, ℓ_2 of randomized one threshold and two threshold classifiers (respectively) satisfying equalized odds and derived from FICO Scores.

cost ratio	$t_{a=Asian,1}$	$P_{a=Asian,1}$	$t_{a=Hispanic,1}$	$P_{a=Hispanic,1}$	$t_{a=Black,1}$	$P_{a=Black,1}$	$t_{a=White,1}$	$P_{a=White,1}$
0.000	0.579	1.000	0.537	1.000	0.475	1.000	0.529	1.000
0.020	0.579	1.000	0.537	1.000	0.475	1.000	0.529	1.000
0.040	1.159	1.000	1.091	1.000	1.002	1.000	1.101	1.000
0.060	3.164	0.999	2.494	1.000	1.496	1.000	2.747	0.999
0.080	3.681	0.999	3.103	1.000	1.989	1.000	3.302	0.999
0.100	4.345	0.999	3.610	1.000	2.492	1.000	3.919	0.998
0.120	5.234	0.998	4.218	0.999	3.007	1.000	4.599	0.998
0.140	6.456	0.996	5.609	0.999	3.991	1.000	5.992	0.996
0.160	7.491	0.994	6.196	0.998	4.504	1.000	6.669	0.994
0.180	7.746	0.994	6.407	0.998	4.982	1.000	7.017	0.994
0.200	11.120	0.989	8.473	0.996	6.000	1.000	9.464	0.990
0.220	12.066	0.987	9.802	0.995	6.497	1.000	10.674	0.988
0.240	12.476	0.985	10.604	0.995	7.014	1.000	11.387	0.986
0.260	13.704	0.982	12.069	0.994	8.009	1.000	12.462	0.983
0.280	14.550	0.980	12.893	0.993	8.994	1.000	13.386	0.980
0.300	15.144	0.978	13.403	0.992	9.523	1.000	13.930	0.979
0.320	15.972	0.976	14.222	0.991	10.498	1.000	14.504	0.976
0.340	19.095	0.969	16.279	0.987	12.491	1.000	16.712	0.967
0.360	20.761	0.966	17.888	0.984	13.991	1.000	18.436	0.961
0.380	23.986	0.963	20.113	0.981	15.506	1.000	20.316	0.953
0.400	26.874	0.963	21.266	0.979	17.475	1.000	22.223	0.946
0.420	29.153	0.965	23.277	0.980	19.502	1.000	24.207	0.941
0.440	32.532	0.975	25.957	0.992	20.494	1.000	27.302	0.939
0.460	32.595	0.975	26.028	0.992	20.517	1.000	27.348	0.939
0.480	33.377	0.978	26.740	0.997	20.995	1.000	28.134	0.939
0.500	33.896	0.980	27.100	1.000	21.197	1.000	28.446	0.939
0.520	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.540	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.560	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.580	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.600	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.620	33.987	0.980	27.150	1.000	21.230	1.000	28.504	0.939
0.640	34.396	0.979	27.466	1.000	21.442	0.997	29.022	0.936
0.660	35.049	0.978	27.977	1.000	21.840	0.993	29.345	0.933
0.680	35.608	0.977	28.483	1.000	22.315	0.989	29.936	0.929
0.700	36.298	0.975	29.012	1.000	22.740	0.984	30.432	0.925
0.720	37.225	0.973	29.565	1.000	23.330	0.978	31.244	0.919
0.740	38.194	0.971	30.481	1.000	23.950	0.971	32.060	0.911
0.760	40.365	0.971	32.023	1.000	24.665	0.958	33.371	0.897
0.780	43.206	0.980	33.532	1.000	25.711	0.945	34.787	0.883
0.800	46.338	0.992	35.531	1.000	27.129	0.928	36.458	0.862
0.820	48.138	0.998	37.012	1.000	28.016	0.918	37.588	0.849
0.840	48.986	1.000	37.362	0.999	28.342	0.913	38.174	0.843
0.860	51.859	1.000	39.296	0.989	29.627	0.891	40.175	0.816
0.880	53.954	1.000	41.492	0.978	31.088	0.869	42.182	0.790
0.900	56.456	1.000	43.418	0.960	32.428	0.840	44.012	0.758
0.920	58.449	1.000	44.410	0.936	33.263	0.813	44.909	0.730
0.940	75.540	1.000	57.908	0.809	42.363	0.607	59.361	0.538
0.960	80.176	1.000	63.658	0.794	46.487	0.564	67.754	0.520
0.980	89.850	1.000	72.362	0.545	56.778	0.407	88.505	0.678
1.000	N/A	0.0	N/A	0.0	N/A	0.0	N/A	0.0

Table 2: Threshold and probability values of a randomized one-threshold classifier satisfying equalized odds derived from FICO scores. (Note: for cost ratio = 1, the probability values are 0 and there is no threshold value because the optimal false positive rate and true positive rate for this cost ratio is 0. Consequently, the classifier returns 0 every time.)

cost ratio	FP_1	FP_2	TP_1	TP_2	ℓ_1/ℓ_2
00.000	0.999	0.999	1.000	1.000	1.000
0.020	0.999	0.999	1.000	1.000	1.000
0.040	0.999	0.999	1.000	1.000	1.000
0.060	0.999	0.999	1.000	1.000	1.000
0.080	0.999	0.999	1.000	1.000	1.000
0.100	0.999	0.999	1.000	1.000	1.000
0.120	0.999	0.999	1.000	1.000	1.000
0.140	0.999	0.999	1.000	1.000	1.000
0.160	0.999	0.999	1.000	1.000	1.000
0.180	0.999	0.999	1.000	1.000	1.000
0.200	0.999	0.999	1.000	1.000	1.000
0.220	0.999	0.999	1.000	1.000	1.000
0.240	0.999	0.999	1.000	1.000	1.000
0.260	0.999	0.999	1.000	1.000	1.000
0.280	0.999	0.999	1.000	1.000	1.000
0.300	0.632	0.631	0.850	0.850	1.000
0.320	0.632	0.631	0.850	0.850	1.000
0.340	0.631	0.631	0.850	0.850	1.000
0.360	0.631	0.631	0.850	0.850	1.000
0.380	0.631	0.631	0.850	0.850	1.000
0.400	0.631	0.631	0.850	0.850	1.000
0.420	0.631	0.631	0.850	0.850	1.000
0.440	0.466	0.466	0.728	0.728	1.000
0.460	0.407	0.407	0.681	0.681	1.000
0.480	0.335	0.334	0.617	0.616	1.000
0.500	0.334	0.334	0.616	0.616	1.000
0.520	0.334	0.334	0.616	0.616	1.000
0.540	0.334	0.334	0.616	0.616	1.000
0.560	0.241	0.241	0.503	0.503	1.000
0.580	0.241	0.241	0.503	0.503	1.000
0.600	0.153	0.153	0.379	0.379	1.000
0.620	0.153	0.153	0.379	0.379	1.000
0.640	0.153	0.153	0.379	0.379	1.000
0.660	0.153	0.153	0.379	0.379	1.000
0.680	0.094	0.093	0.252	0.251	1.000
0.700	0.035	0.032	0.121	0.114	1.000
0.720	0.035	0.032	0.121	0.114	1.001
0.740	0.035	0.032	0.121	0.114	1.002
0.760	0.035	0.032	0.121	0.114	1.003
0.780	0.000	0.032	0.000	0.114	1.000
0.800	0.000	0.000	0.000	0.000	1.000
0.820	0.000	0.000	0.000	0.000	1.000
0.840	0.000	0.000	0.000	0.000	1.000
0.860	0.000	0.000	0.000	0.000	1.000
0.880	0.000	0.000	0.000	0.000	1.000
0.900	0.000	0.000	0.000	0.000	1.000
0.920	0.000	0.000	0.000	0.000	1.000
0.940	0.000	0.000	0.000	0.000	1.000
0.960	0.000	0.000	0.000	0.000	1.000
0.980	0.000	0.000	0.000	0.000	1.000
1.000	0.000	0.000	0.000	0.000	1.000

Table 3: False positive rates, FP_1, FP_2 and true positive rates, TP_1, TP_2 and ratio of the losses ℓ_1, ℓ_2 of randomized one threshold and two threshold classifiers (respectively) satisfying equalized odds derived from COMPAS Scores.

cost ratio	$t_{a=White,1}$	$P_{a=White,1}$	$t_{a=Black,1}$	$P_{a=Black,1}$
0.000	1.003	1.000	1.006	1.000
0.020	1.003	1.000	1.006	1.000
0.040	1.003	1.000	1.006	1.000
0.060	1.003	1.000	1.006	1.000
0.080	1.003	1.000	1.006	1.000
0.100	1.003	1.000	1.006	1.000
0.120	1.003	1.000	1.006	1.000
0.140	1.003	1.000	1.006	1.000
0.160	1.003	1.000	1.006	1.000
0.180	1.003	1.000	1.006	1.000
0.200	1.003	1.000	1.006	1.000
0.220	1.003	1.000	1.006	1.000
0.240	1.003	1.000	1.006	1.000
0.260	1.003	1.000	1.006	1.000
0.280	1.003	1.000	1.006	1.000
0.300	1.998	1.000	3.168	0.977
0.320	1.998	1.000	3.168	0.977
0.340	2.001	1.000	3.182	0.977
0.360	2.001	1.000	3.182	0.977
0.380	2.001	1.000	3.182	0.977
0.400	2.001	1.000	3.182	0.977
0.420	2.001	1.000	3.182	0.977
0.440	3.000	1.000	4.731	0.990
0.460	3.485	1.000	5.307	1.000
0.480	4.028	0.982	5.994	1.000
0.500	4.038	0.982	6.004	1.000
0.520	4.038	0.982	6.004	1.000
0.540	4.038	0.982	6.004	1.000
0.560	4.770	0.936	7.000	1.000
0.580	4.770	0.936	7.000	1.000
0.600	5.594	0.854	8.000	1.000
0.620	5.594	0.854	8.000	1.000
0.640	5.594	0.854	8.000	1.000
0.660	5.594	0.854	8.000	1.000
0.680	5.896	0.616	8.989	1.000
0.700	7.530	0.508	9.950	0.999
0.720	7.530	0.508	9.950	0.999
0.740	7.530	0.508	9.950	0.999
0.760	7.530	0.508	9.950	0.999
0.8 – 1.0	N/A	0.0	N/A	0.0

Table 4: Threshold and probability values of a randomized one-threshold classifier satisfying equalized odds derived from COMPAS scores. (Note: for cost ratio $\in [0.780, 1]$, the probability values are 0 and there is no threshold value because the optimal false positive rate and true positive rate for this cost ratio is 0. Consequently, the classifier returns 0 every time.)